



Continuous Assessment Test (CAT) – I AUGUST 2025

Programme	:	BTech CSE with Specialization in Artificial Intelligence and Machine Learning	Semester	:	Fall 2025
Course Code & Course Title	:	BCSE419L Speech and Language Processing	Class Number	:	CH2025260102057 CH2025260102059
Faculty	:	Dr G Bharadwaja Kumar Dr M Janaki Meena	Slot	:	F1 + TF1
Duration	:	90 minutes	Max. Mark	:	50

General Instructions:

Write only your registration number on the question paper in the box provided and do not write other information

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level												
1		Given a small labelled training set of five short movie-review snippets and one test sentence. Training Set <table><tr><th>Class</th><th>Document</th></tr><tr><td>-</td><td>boring and predictable</td></tr><tr><td>-</td><td>waste of time and money</td></tr><tr><td>+</td><td>absolutely amazing and thrilling</td></tr><tr><td>+</td><td>wonderful performances and great story</td></tr><tr><td>-</td><td>dull script and bad acting</td></tr></table> Test Sentence boring but amazing story Classify the test sentence as either Positive (+) or Negative (-) using a multinomial Naïve Bayes classifier with add-one smoothing.	Class	Document	-	boring and predictable	-	waste of time and money	+	absolutely amazing and thrilling	+	wonderful performances and great story	-	dull script and bad acting	10	2	3
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2.	a.	The Skip-gram model in Word2Vec predicts surrounding context words given a target word within a specified window size. Suppose you are training a Skip-gram model with: Window size = 2 Vocabulary size = 10,000 Embedding dimension = 300 Training corpus size = 50 million tokens Consider the sentence “College life is like a Tom and Jerry episode” with Tom as target word and build a sample training data for the skip gram model with negative sampling.	7	2	3												
	b.	You observe that the embeddings learned by the skip gram model in section a,	3	2	4												

		capture strong syntactic relationships (e.g., run ↔ running), but weaker semantic relationships (e.g., king ↔ queen). Explain how the choice of window size in Skip-gram affects the balance between capturing syntactic vs. semantic relationships.			
3.		You are building a bigram language model for an email autocomplete system. Your training corpus is relatively small (50,000 sentences), so many bigrams in user queries are unseen in training. During evaluation on a test set, you observe: Several grammatically correct sentences receive probability = 0, making perplexity undefined. Propose two smoothing techniques to handle unseen bigrams in this system. Briefly describe how each one redistributes probability mass by borrowing probability mass from lower-order n-grams or by considering the diversity of contexts in which a word appears.	10	2	3
4.	a.	Given the following context-free grammar, parse the sentence “the cat sat on the mat” S → NP VP VP → V NP VP → V NP PP PP → P NP NP → Det N NP → NP PP Det → "the" N → “cat” N → “mat” V → “sat” P → “on”	5	1	3
	b.	Construct dependency tree for the sentence “My mother gave me a new bike for my birthday.”	5	1	3
5.		Given the following list of words, apply the Porter Stemming process step-by-step to obtain their stems: - studies - mice - geese - relational - unhappiness - impurities What will be output given by Lemmatization technique for the above words? What is the advantage Lemmatization technique over the stemming technique?	10	1	3

*****All the best *****